

PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement #38 : Dropshipping Inventory & Order
Orchestrator

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SECTION : I

Use Case: Receive and Process Order Webhook

Use Case ID: UC-01 Primary Actor: Storefront Platform

1. Preconditions

- The Storefront Platform is configured to send order webhooks to the system.
- The system is running and available to receive incoming webhooks.
- Supplier inventory information is accessible through the supplier API.

2. Postconditions

- The order is successfully processed and supplier purchase order(s) are created for available products.
- Multi-vendor products are separated into the appropriate supplier orders when required.
- The supplier order status is updated in the system.

3. Main Success Scenario

1. The Storefront Platform sends a new customer order webhook to the system.
2. The system receives and validates the incoming webhook.
3. The system parses the order and identifies the ordered SKUs.
4. The system identifies the supplier associated with each SKU.
5. The system requests inventory availability from the relevant Supplier Partner.
6. The Supplier Partner returns the inventory availability information.
7. The system verifies that the required SKUs are available.
8. If products belong to multiple suppliers, the system splits the order into supplier-specific orders.
9. The system creates supplier purchase order(s).
10. The system sends the purchase order(s) to the relevant Supplier Partner(s).
11. The system updates the supplier order status.
12. The system confirms successful processing of the order.

4. Alternate Flow – Supplier SKU Out of Stock

1. The Supplier Partner reports that one or more required SKUs are unavailable.
2. The system marks the affected SKU as unavailable.
3. The system does not create a supplier purchase order for the unavailable SKU.
4. The system notifies the E-Store Owner about the inventory issue.
5. The system updates the order status to indicate the inventory problem.
6. The use case ends with the order requiring further action